GaussianNB

class sklearn.naive_bayes.**GaussianNB**(*, *priors*=*None*, *var_smoothing*=*1e-09*)

Gaussian Naive Bayes (GaussianNB). [source]

Can perform online updates to model parameters via `partial_fit`. For details on algorithm used to update feature means and variance online, see Stanford CS tech report STAN-CS-79-773 by Chan, Golub, and LeVeque.

Read more in the User Guide.

Parameters:

priors : *array-like of shape (n_classes,), default=None*

Prior probabilities of the classes. If specified, the priors are not adjusted according to the data.

var_smoothing : *float, default=1e-9*

Portion of the largest variance of all features that is added to variances for calculation stability.

> *Added in version 0.20.*

Attributes:

class_count_ : *ndarray of shape (n_classes,)*

number of training samples observed in each class.

class_prior_ : *ndarray of shape (n_classes,)*

probability of each class.

classes_ : *ndarray of shape (n_classes,)*

class labels known to the classifier.

epsilon_ : *float*

absolute additive value to variances.

n_features_in_ : *int*

Number of features seen during fit.

Added in version 0.24.

feature_names_in_ : *ndarray of shape (* `n_features_in_` *,)*

Names of features seen during fit. Defined only when `X` has feature names that are all strings.

Added in version 1.0.

var_ : *ndarray of shape (n_classes, n_features)*

Variance of each feature per class.

Added in version 1.0.

theta_ : *ndarray of shape (n_classes, n_features)*

mean of each feature per class.

See also

`BernoulliNB`

Naive Bayes classifier for multivariate Bernoulli models.

`CategoricalNB`

Naive Bayes classifier for categorical features.

`ComplementNB`

Complement Naive Bayes classifier.

`MultinomialNB`

Naive Bayes classifier for multinomial models.

Examples

```
>>> import numpy as np
>>> X = np.array([[-1, -1], [-2, -1], [-3, -2], [1, 1], [2, 1], [3, 2]])
>>> Y = np.array([1, 1, 1, 2, 2, 2])
>>> from sklearn.naive_bayes import GaussianNB
>>> clf = GaussianNB()
>>> clf.fit(X, Y)
GaussianNB()
>>> print(clf.predict([[-0.8, -1]]))
[1]
```

```
>>> clf_pf = GaussianNB()
>>> clf_pf.partial_fit(X, Y, np.unique(Y))
GaussianNB()
>>> print(clf_pf.predict([[-0.8, -1]]))
[1]
```

`fit(X, y, sample_weight=None)` [source]

Fit Gaussian Naive Bayes according to X, y.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

Training vectors, where `n_samples` is the number of samples and `n_features` is the number of features.

y : ***array-like of shape (n_samples,)***

Target values.

sample_weight : ***array-like of shape (n_samples,), default=None***

Weights applied to individual samples (1. for unweighted).

> ***Added in version 0.17:*** Gaussian Naive Bayes supports fitting with *sample_weight*.

Returns:

self : ***object***

Returns the instance itself.

`get_metadata_routing()` [source]

Get metadata routing of this object.

Please check User Guide on how the routing mechanism works.

Returns:

routing : ***MetadataRequest***

A `MetadataRequest` encapsulating routing information.

`get_params(deep=True)` [source]

Get parameters for this estimator.

Parameters:

deep : ***bool, default=True***

If True, will return the parameters for this estimator and contained subobjects that are estimators.

Returns:

params : ***dict***

Parameter names mapped to their values.

`partial_fit(X, y, classes=None, sample_weight=None)` [source]

Incremental fit on a batch of samples.

This method is expected to be called several times consecutively on different chunks of a dataset so as to implement out-of-core or online learning.

This is especially useful when the whole dataset is too big to fit in memory at once.

This method has some performance and numerical stability overhead, hence it is better to call partial_fit on chunks of data that are as large as possible (as long as fitting in the memory budget) to hide the overhead.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

Training vectors, where `n_samples` is the number of samples and `n_features` is the number of features.

y : ***array-like of shape (n_samples,)***

Target values.

classes : ***array-like of shape (n_classes,), default=None***

List of all the classes that can possibly appear in the y vector.

Must be provided at the first call to partial_fit, can be omitted in subsequent calls.

sample_weight : ***array-like of shape (n_samples,), default=None***

Weights applied to individual samples (1. for unweighted).

> ***Added in version 0.17.***

Returns:

self : ***object***

Returns the instance itself.

`predict(X)` [source]

Perform classification on an array of test vectors X.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

The input samples.

Returns:

C : ***ndarray of shape (n_samples,)***

Predicted target values for X.

`predict_joint_log_proba(X)` [source]

Return joint log probability estimates for the test vector X.

For each row x of X and class y, the joint log probability is given by `log P(x, y) = log P(y) + log P(x|y),` where `log P(y)` is the class prior probability and `log P(x|y)` is the class-conditional probability.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

The input samples.

Returns:

C : ***ndarray of shape (n_samples, n_classes)***

Returns the joint log-probability of the samples for each class in the model. The columns correspond to the classes in sorted order, as they appear in the attribute classes_.

`predict_log_proba(X)` [source]

Return log-probability estimates for the test vector X.

Parameters:

Parameters:

X : ***array-like of shape (n_samples, n_features)***

The input samples.

Returns:

C : ***array-like of shape (n_samples, n_classes)***

Returns the log-probability of the samples for each class in the model. The columns correspond to the classes in sorted order, as they appear in the attribute classes_.

`predict_proba(X)` [source]

Return probability estimates for the test vector X.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

The input samples.

Returns:

C : ***array-like of shape (n_samples, n_classes)***

Returns the probability of the samples for each class in the model. The columns correspond to the classes in sorted order, as they appear in the attribute classes_.

`score(X, y, sample_weight=None)` [source]

Return accuracy on provided data and labels.

In multi-label classification, this is the subset accuracy which is a harsh metric since you require for each sample that each label set be correctly predicted.

Parameters:

X : ***array-like of shape (n_samples, n_features)***

Test samples.

y : ***array-like of shape (n_samples,) or (n_samples, n_outputs)***

True labels for `X`.

sample_weight : ***array-like of shape (n_samples,), default=None***

Sample weights.

Returns:

Returns:

score : *float*

Mean accuracy of `self.predict(X)` w.r.t. `y` .

set_fit_request(*, *sample_weight: bool | None | str = '$UNCHANGED$'*) → GaussianNB [source]

Configure whether metadata should be requested to be passed to the `fit` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True` : metadata is requested, and passed to `fit` if provided. The request is ignored if metadata is not provided.
- `False` : metadata is not requested and the meta-estimator will not pass it to `fit` .
- `None` : metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str` : metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

> ! *Added in version 1.3.*

Parameters:

sample_weight : *str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*

Metadata routing for `sample_weight` parameter in `fit` .

Returns:

self : *object*

The updated object.

set_params(***params*) [source]

Set the parameters of this estimator.

The method works on simple estimators as well as on nested objects (such as `Pipeline`). The latter have parameters of the form `<component>__<parameter>` so that it's possible to update each component of a nested object.

Parameters:

****params : *dict***

Estimator parameters.

Returns:

self : *estimator instance*

Estimator instance.

set_partial_fit_request(**, classes: bool | None | str = '$UNCHANGED$', sample_weight: bool | None | str = '$UNCHANGED$'*) → GaussianNB [source]

Configure whether metadata should be requested to be passed to the `partial_fit` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True` : metadata is requested, and passed to `partial_fit` if provided. The request is ignored if metadata is not provided.
- `False` : metadata is not requested and the meta-estimator will not pass it to `partial_fit`.
- `None` : metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str` : metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

Parameters:

classes : *str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*

Metadata routing for `classes` parameter in `partial_fit`.

sample_weight : *str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*

Metadata routing for `sample_weight` parameter in `partial_fit`.

Returns:

self : *object*

The updated object.

`set_score_request(*, sample_weight: bool | None | str = '$UNCHANGED$')` → GaussianNB [source]

Configure whether metadata should be requested to be passed to the `score` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True` : metadata is requested, and passed to `score` if provided. The request is ignored if metadata is not provided.
- `False` : metadata is not requested and the meta-estimator will not pass it to `score`.
- `None` : metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str` : metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

> ***Added in version 1.3.***

Parameters:

sample_weight : ***str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED***

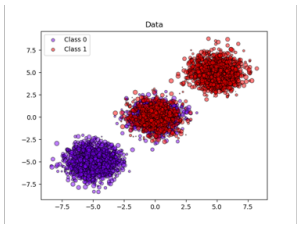
Metadata routing for `sample_weight` parameter in `score`.

Returns:

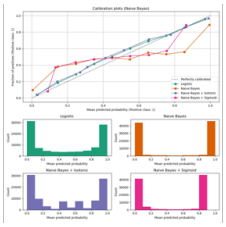
self : ***object***

The updated object.

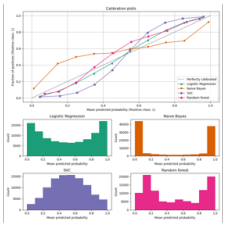
Gallery examples



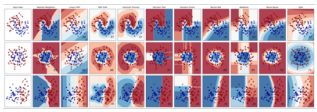
Probability calibration of classifiers



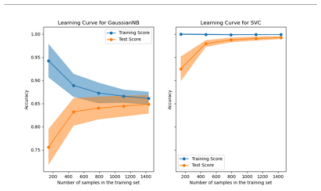
Probability Calibration curves



Comparison of Calibration of Classifiers



Classifier comparison



Plotting Learning Curves and Checking Models' Scalability

Previous
ComplementNB

Next
MultinomialNB

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